

**Operator UI / TUI + twin renderer**  
UI → API writes; renderer displays state

**REST / WebSocket + A2A**  
shared authenticated entry points

**MCP interface**  
read-oriented subset; not all host tools

**PLATFORM PROCESS**

**Agent orchestrator**

harness adapters · team channels  
lead / workers · tasks · memory  
host tools + scoped dispatch  
shared node / team / run IDs

intent

**Binding + approval bridge**

node semantics · read/write scope  
manual / auto policy · approvals  
agent-write admission + readback  
records · separate backstop

write

**Industrial supervisor**

DAQ runtimes + gateway sweep  
DCW dispatch + protocol drivers  
alarms · line / recipe / run state  
device-twin state publication

**External CLI engines**

separate processes; containment external

**Separate persistent stores**

platform DB · time series · objects · audit

**Plant / protocol boundary**

physical devices or seeded scenarios

Logical IDs; not atomic storage

Solid: request / control · Dashed: observation / record flow · PLC safety and hard real-time control remain external.